

Trace elements in modern and archaeological human teeth: Implications for human metal exposure and enamel diagenetic changes



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ABSTRACT

The chemical composition of human tooth enamel from individuals born at different times and/or different geographical places can provide a record of human metal exposure through time and space. However, in addition to “in-vivo” metal exposure, post-mortem diagenetic changes can also affect the trace element composition of enamel from archaeological contexts. Modern human enamel collected from individuals born during the 20th century exhibit elevated concentrations of metals used in the industry such as Cr, Ni, Cu, Zn, and Pb when compared to archaeological enamel from disparate localities. This is a result of overall higher human exposure to industrial metals at present compared to pre-industrial times. In contrast, archaeological enamel tends to exhibit lower trace metal concentrations, however, some archaeological samples show elevated concentrations of V, Mn, Fe, REE, Th, and U relative to modern enamel. The elevated concentrations of these metals indicate post-mortem addition of elements into the preserved enamel matrix. Assuming the highest “in-vivo” human exposure to industrial metals occurred at present, we calculated maximum threshold concentrations (MTC) based on data derived from modern human enamel. Archaeological enamel samples showing concentrations for one or more of the following elements, V, Mn, Fe, REE, Th, and/or U, higher than the MTC strongly indicate modification of their trace metal compositions during the post-mortem interval. Normalization of the archaeological enamel samples concentrations for V, Mn, Fe, La, Ce, Nd, Dy, Yb, Th, and U to the MTC for these metals provides a rapid method for screening samples for diagenetic alteration.

Examples of MTC evaluation of samples from several diverse archaeological contexts show that even in locations with similar age and geological context the preservation state of archaeological enamel can range from pristine, likely preserving the “in-vivo” biogenic chemical signal, to slightly altered, as a result of adsorption of redox sensitive metals, to highly diagenetically altered, apparently accompanied with cation substitution in the apatite structure. Mass-balance considerations show that the slightly altered enamel samples should still preserve the original “in-vivo” Sr isotope signal, but the Pb isotope signal may be compromised. In contrast, in the highly altered enamel samples the “in-vivo” chemical signal for a number of elements, including Sr and Pb isotopes, may well be affected by the diagenetic signal. This work demonstrates how trace element analyses can enhance isotopic analysis on tooth enamel samples from all archaeological contexts in order to evaluate the alteration status of the sample(s) under study. Such trace element screening will allow researchers to select the best preserved human enamel samples for isotopic and concentration information to decipher nutritional status, metal exposure, and geographical place of origin in past populations.

1. Introduction

In humans, tooth enamel mineralization for the permanent dentition occurs during early childhood, from about 3 to 4 months to 12–16 years age, depending on the developmental timing of the tooth (e.g., Reid and

Dean, 2006). Once enamel formation is complete and the enamel fully mineralized, childhood elemental and isotope signals incorporated within the enamel matrix will remain unchanged throughout life (e.g., Gulson and Gillings, 1997). Therefore, the elemental concentration and isotope ratio information recorded in modern and archaeological

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human enamel can provide information about nutritional status, metal exposure, and geographical place of origin (e.g., Turner et al., 2009; Austin et al., 2013; Reynard and Balter, 2014; Kamenov and Gulson, 2014; Valentine et al., 2015; Price et al., 2017).

A major complication of using the chemical information recorded in archaeological skeletal material is the possibility of post-mortem contamination from the burial environment. This process of diagenetic change is well-documented for preserved animal bone (e.g., Nelson et al., 1986; Trueman et al., 2004; Reynard and Balter, 2014; Keenan and Engel, 2017). Bones are characteristically porous with relatively high organic content and are prone to rapid diagenetic alteration. Studies have demonstrated rapid (months to years) structural and chemical transformation of animal bones during diagenesis (Trueman et al., 2004; Keenan and Engel, 2017). As a result, the chemical and isotope signal in archaeological bone may not be a good proxy to decipher nutrition status, metal exposure, and geographical place of origin as it is a mixture of the “in-vivo” signal and elements added and/or removed during the post-mortem interval (Kohn and Moses, 2013; Reynard and Balter, 2014).

In contrast to bones, tooth enamel has high degree of mineralization and larger bioapatite crystallites and, generally, is believed to be resistant to diagenetic alterations and thus preserves the biogenic elemental and isotope signal (e.g., Lee-Thorp and van der Merwe, 1991; Ayliffe et al., 1994; MacFadden et al., 2010). As a result, archaeological tooth enamel is the material of choice for determination of C, O, Sr and Pb isotopic compositions of past populations (e.g., Turner et al., 2009; Valentine et al., 2015; Price et al., 2017). Although tooth enamel is better than bone to analyze for pre-historic populations in circumstances where bone preservation is poor, it may also be compromised depending upon the vagaries of burial. For example, Chiaradia et al. (2003) demonstrated that in burial sites with loose, coarse-grained soils pre-historic human teeth may be contaminated by modern anthropogenic Pb. Recent work by Giovas et al. (2016) also showed that zooarchaeological tooth enamel may be contaminated by modern anthropogenic Pb in the burial environment. Apparently, meteoric water can infiltrate and transport soluble Pb compounds to deeper archaeological horizons and become incorporated into tissue, including archaeological human and animal tooth enamel (Chiaradia et al., 2003; Giovas et al., 2016). In addition to post-mortem Pb contamination, a number of other metals also can be transported in the ground water of the burial environment and could potentially alter the chemical composition of the tooth enamel. This effect is well-documented in the paleontological literature. For example, Kohn et al. (1999) demonstrated a marked increase in Fe and Mn in 3.9 Ma-year-old fossil animal teeth. Similarly, Hinz and Kohn (2010) showed that patterns of trace element distributions in 25 ka-year-old animal enamel also reflect diagenetic alteration. It is not clear, however, if similar diagenetic effects can be observed in human enamel at much shorter (hundreds to thousand years) post-mortem intervals.

The contemporary scientific literature is abound with human enamel isotope works based on the assumption that human tooth enamel is resistant to diagenetic alteration and preserves the biogenic isotope signal. Countless studies use archaeological human enamel as a proxy for biogenic isotope signal (e.g., Turner et al., 2009; Austin et al., 2013; Scharlotta and Weber, 2014; Valentine et al., 2015; Price et al., 2017; and references therein), but such studies lack methods to test if the enamel has undergone diagenetic alteration during the post-mortem interval. Detailed investigation of the trace element composition of modern and archaeological human enamel can provide a simple method to distinguish pristine from diagenetically altered samples. In this work, we compare the elemental composition of modern human enamel samples with select archaeological human enamel samples to assess human metal exposure through time and identify metal markers indicating post-mortem diagenetic alteration.

2. Samples and analytical methods

All reagents used in this work were Optima-grade (www.fishersci.com) and sample preparation was performed at the University of Florida in the Department of Anthropology's Bone Chemistry Laboratory and in the Department of Geological Sciences' Clean Laboratory (class 500, equipped with class 10 laminar flow hoods). Modern teeth were collected following standard dental extraction procedures from individuals from Europe, North America, Central America, South America, Caribbean, and Africa ($n = 77$, on-line supplement). Archaeological samples of human teeth were recovered from pre-historic and early historic Florida sites, the Philippines, and from an early historic site in Peru (Lofaro, 2016) (on-line supplement).

A 20–40 mg chunk of enamel was carefully cleaned and dentine was removed mechanically using a dental drill and microscope apparatus. All external surfaces were abraded until only clean enamel, free of any dentine or visible surficial contamination, remained. The clean enamel was weighed again and dissolved in 8N HNO₃ in acid-washed closed Teflon vials for 24h on a hot plate at 120 °C. Following complete dissolution the vials were opened and the sample solution was evaporated to dryness. Dilute nitric acid (0.8 N HNO₃), spiked with 8 ppb Rh and Re was added to the dry residue to achieve final dilution around 2000x. The exact final dilution for elemental analyses was determined by weight for each sample. Trace element analyses were performed on an Element2 HR-ICP-MS in medium resolution with Re and Rh used as internal standards. The elements from Mg to Ba were normalized to ¹⁰³Rh, and the elements from La to U were normalized to ¹⁸⁵Re. Quantification of the results was achieved via external calibration using a set of gravimetrically prepared dilutions of commercial ICP-MS standards (SPEX CertiPrep, Inc.). Limit of detection for each element analyzed in this work are provided in the on-line supplement. Samples of Bone Ash SRM 1400 standard were prepared and analyzed in concert with the samples to evaluate accuracy and precision of the analyses of the unknown samples (on-line supplement). Based on reproducibility observed for SRM 1400 ($n = 16$) analyzed together with the samples the reported concentration values are better than $\pm 5\%$ for Mg, Ca, V, Cr, Fe, Zn, Sr, Ba, La, Ce, Pr, Nd, Pb, and U, and better than $\pm 10\%$ for Mn, Sm, Dy (on-line supplement). Several elements showed higher than 10% reproducibility in SRM 1400, including Ni (19%), Cu (13%), Eu (24%), Gd (25%), Tb (11%), Ho (18%), Tm (16%), Yb (12%), Lu (14%), and Th (18%) (on-line supplement).

3. Results

Detailed major and trace element data for modern and archaeological enamel samples are available in the on-line supplement. Table 1 summarizes the average, standard deviation, minimum, and maximum concentrations measured in the modern and archaeological human enamel samples analyzed in the University of Florida following the procedures described in the previous section. The two major metals, Mg and Ca, although slightly elevated on average in the modern enamel, overall show similar concentrations to the archaeological enamel samples when considering analytical error (Table 1). The average concentration in the modern enamel (39.1%, $\pm 1.8\%$) is within error of the average Ca concentration in the archaeological enamel (Ca = 37.7% ($\pm 1.4\%$)). Similarly, the average Mg concentration in the modern enamel (Mg = 3075 ppm, ± 623 ppm) is also within error with the archaeological enamel (Mg = 2430 ppm, ± 293 ppm) (Table 1). Sr and Ba are also similar in modern and archaeological enamel (Table 1).

Modern enamel shows overall elevated Cr, Ni, Cu, Zn, and Pb concentrations compared to the archaeological enamel. Ni and Cr in the archaeological enamel samples analyzed in this work are always below 1 ppm (Table 1). In contrast, a number of modern samples show Cr and Ni above 1 ppm, with maximum 10.3 ppm for Cr and 36.5 ppm for Ni (Table 1). The modern samples show average Pb = 6.55 ppm, about an order of magnitude higher compared to the archaeological samples

Table 1

Average, standard deviation, minimum, maximum concentrations (ppm) observed in modern and archaeological enamel samples. Maximum Threshold Concentrations (MTC): $MTC = \text{Modern Max} + 2 \times \text{Modern STDEV}$ are calculated from the modern human enamel samples.

	MODERN ENAMEL				^a MTC	ARCHAEOLOGICAL ENAMEL			
	Average	STDEV	Min	Max		Average	STDEV	Min	Max
Mg	3075	623	1931	5225	6472	2430	293	1743	3082
Ca	391417	18230	350011	425928	462387	377112	14049	347506	401221
V	0.028	0.014	bdl	0.079	0.11	0.190	0.177	bdl	0.654
Cr	0.8	1.83	bdl	10.3	13.9	0.072	0.112	0.020	0.609
Mn	1.95	1.72	0.51	12	15.4	12.70	23.60	bdl	152
Fe	15	19	2.23	104	143	23	66	2.8	447
Ni	2.2	4.9	0.11	36.5	46	0.24	0.17	0.03	0.91
Cu	3.6	7.4	0.01	32.9	48	0.25	0.34	bdl	1.44
Zn	215	77	80.4	474	629	145	98	51	687
Sr	168	178	58	1344	1700	188	45	84	373
Ba	4.6	4.4	0.52	31.3	40	3.2	3.9	0.45	19.6
La	0.007	0.012	bdl	0.079	0.1	0.057	0.269	0.002	1.820
Ce	0.010	0.015	bdl	0.087	0.12	0.095	0.378	0.003	2.554
Pr	0.0014	0.002	bdl	0.012	0.015	0.012	0.056	bdl	0.381
Nd	0.006	0.008	bdl	0.042	0.058	0.051	0.217	bdl	1.46
Sm	0.004	0.005	bdl	0.020	0.030	0.013	0.047	bdl	0.279
Eu	bdl	bdl	bdl	bdl	-	0.0060	0.0140	bdl	0.0571
Gd	bdl	bdl	bdl	bdl	-	0.0260	0.0650	bdl	0.2675
Tb	0.0005	0.0003	bdl	0.0017	0.002	0.0020	0.006	bdl	0.0391
Dy	0.0015	0.0015	bdl	0.0056	0.009	0.0095	0.037	bdl	0.2374
Ho	0.0005	0.0004	bdl	0.0012	0.002	0.003	0.008	bdl	0.046
Er	0.0014	0.0013	bdl	0.0057	0.008	0.007	0.025	bdl	0.140
Tm	0.0003	0.0002	bdl	0.0007	0.001	0.002	0.005	bdl	0.019
Yb	0.0026	0.0031	bdl	0.0148	0.021	0.006	0.022	bdl	0.120
Lu	0.0005	0.0004	bdl	0.0017	0.003	0.002	0.004	bdl	0.015
Pb	6.55	8.6	0.04	45.5	63	0.63	1.41	bdl	7.20
Th	0.0031	0.006	bdl	0.035	0.05	0.006	0.012	bdl	0.078
U	0.003	0.005	bdl	0.036	0.05	0.011	0.023	0.001	0.155

^a MTC - Maximum Threshold Concentration.

(average Pb = 0.63 ppm, Table 1). Overall, the majority of the archaeological samples (n = 38) show sub-ppm Pb content, and 7 samples show Pb above 1 ppm (on-line supplement). Modern sample F04 shows very high Cu (1107 ppm) and Zn (650 ppm), likely a result of dental restoration (on-line supplement), and these two measurements are excluded from the average, minimum, and maximum Cu and Zn presented in Table 1. Even excluding sample F04, Cu is on average over an order of magnitude elevated in the modern samples when compared to the archaeological enamel (Table 1). Zn is lower on the low concentration end (min = 51 ppm) in the archaeological enamel when compared to the modern enamel low end (min = 80 ppm) (Table 1). The average Zn concentrations are overall elevated in the modern enamel with exception of archaeological sample 2908_1. Zn in this archaeological sample is 687 ppm and although similar to the maximum observed in the modern F04 outlier, it shows also a number of other elements like V, Mn, Fe, REE, and U with highly elevated concentrations.

V, Mn, Fe, REE, Th and U show elevated concentrations in some of the archaeological enamel samples. The average and maximum V, Mn, and Fe concentrations are overall higher in the archaeological enamel analyzed in this work compared to the modern enamel (Table 1). REE, Th and U on the low-end (minimum values) of the archaeological samples are similar to modern enamel and are mostly below detection limit (Table 1). On the high-end, however, some archaeological samples, in particular sample 2908_1, show elevated REE concentrations (on-line supplement, Fig. 1).

Maximum Threshold Concentration (MTC) for each element presented in Table 1 is calculated from the 77 modern enamel samples analyzed in this work (on-line supplement). MTC is calculated using the observed modern enamel maximum concentration plus 2 times the STDEV for each element. Concentrations (C) of V, Mn, Fe, La, Ce, Nd, Dy, Yb, Th, and U in three selected archaeological enamel samples normalized to MTC (C/MTC) are plotted in Fig. 1. All three

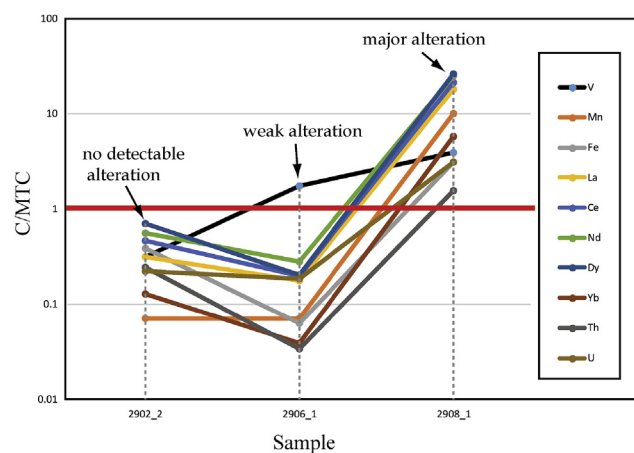


Fig. 1. Three pre-historic samples (Woodland Period) from Florida Native American human enamel normalized to V, Mn, Fe, La, Ce, Nd, Dy, Yb, Th, and U Maximum Threshold Concentrations (MTC) listed in Table 1. Note that all three samples are from very similar burial context. Sample 2902_2 is from Palmetto Mound, and samples 2906_1 and 2908_1 are from Hughes Island Mound, both sites are located along the west coast of North-Central Florida and share similar geological context. Sample 2902_2 shows concentrations for all elements below the MTC ($C/MTC < 1$) indicating no detectable alteration in this enamel sample. Sample 2906_1 shows only V above the MTC, indicating weak alteration. In contrast, sample 2908_1 shows all elements above the MTC indicating major alteration. Elemental and isotope data from 2908_1 should not be used for dietary or migratory studies as the chemical composition of this sample is significantly changed due to addition of elements in the burial environment.

archaeological enamel samples plotted on Fig. 1 are from similar burial context (Weeden Island) dating to the Middle Woodland Period in Florida. Sample 2902_2 is from Palmetto Mound and samples 2906_1 and 2908_1 are from Hughes Island Mound. Both sites are located on

the Gulf Coast of North-Central Florida with the same geological context. As can be seen from the figure, sample 2902_2 shows C/MTC for all of the above elements below 1 (the red line on Fig. 1). This indicates that in this archaeological enamel sample the metal concentrations are within the possible “in-vivo” range and therefore are not detectably enhanced during the post-mortem interval. This sample most likely preserves the “in-vivo” trace element signal and the chemical and isotope information can be used for anthropological interpretations. The second sample, 2906_1, labeled as “weak alteration”, shows only V above the MTC, suggesting that in this sample V was slightly enhanced during the diagenesis. The third sample, 2908_1, enamel data show C/MTC well above the MTC of 1, indicating that the addition of metals during diagenesis significantly altered the “in-vivo” signal. The chemical composition of such a sample is a mixture of “in-vivo” and diagenetic signals and cannot (and should not) be used for anthropological interpretations, as will be discussed below.

4. Discussion

4.1. Enamel elemental composition and human metal exposure

The “in-vivo” chemical composition of human enamel is controlled by its crystal structure, human physiology and nutrition, and exposure to metals during childhood. According to the World Health Organization, the metals analyzed in this work belong to four different groups in terms of human physiology and nutrition (<http://www.who.int/nutrition>). Ca and Mg are major nutrients, Fe, Cu, Zn, and Cr are essential trace elements, V, Mn, Ni, Sr, and Ba are ‘probable’ essential trace elements, and REE (rare earth elements), Pb, Th, and U are non-essential elements. Ca concentration in the enamel is controlled by the crystal structure of hydroxyapatite ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) and Mg and CO_3^{2-} are considered the main impurities (e.g., Trueman and Tuross, 2002). Comparison of modern and archaeological enamel shows that Ca and Mg show overall similar concentrations (Table 1), a result of the apatite crystal structure and the essential nutrition status of these two metals.

Trace element abundances in apatite also can be controlled by the crystal structure via charge and size substitutions. As a result, natural inorganic apatites typically contain elements such as U, Th, and REE on the order of tens to thousands of ppm (e.g., Pan and Fleet, 2002). In contrast, modern human enamel shows well below 0.1 ppm concentrations for U, Th, and REE (Table 1). This indicates that human physiology plays a much more important role for the “in-vivo” U, Th, and REE enamel composition than the crystal structure. In addition to physiology and nutrition, environmental exposure to certain metals during the formation of the enamel also plays a role in the observed trace element abundances in the human enamel. For example, anthropogenic Pb becomes incorporated in human enamel once Pb mining and usage begins in a given region (Kamenov and Gulson, 2014). Work by Budd et al. (2004) and Montgomery et al. (2010) showed overall lower Pb concentrations in enamel from prehistoric populations (0.003–0.68 ppm) compared to Romano-British and Medieval populations (0.02–30.1 ppm), a result of Pb mining and increased use of Pb in Europe during and after the Roman Empire. Similarly, elevated Pb was observed in archaeological enamel from pre-industrial mining site inhabitants, also due to enhanced Pb exposure (Price et al., 2017).

Although Pb, and metals like Fe and Cu, were mined in the archaeological past, most mining activities were localized and relatively minor compared to modern industrial times. Human activities at present have greatly increased the circulation of metals in the environment (air, water, and soils) and these metals inevitably enter the human food chain (Nriagu and Pacyna, 1988). At present, the global environmental cycles of virtually all industrial metals are dominated by anthropogenic activities (Nriagu and Pacyna, 1988; Rauch and Pacyna, 2009). Studies of natural archives of environmental change through time, such as peat, lake, and marine sediments (Fig. 2), clearly demonstrate highly

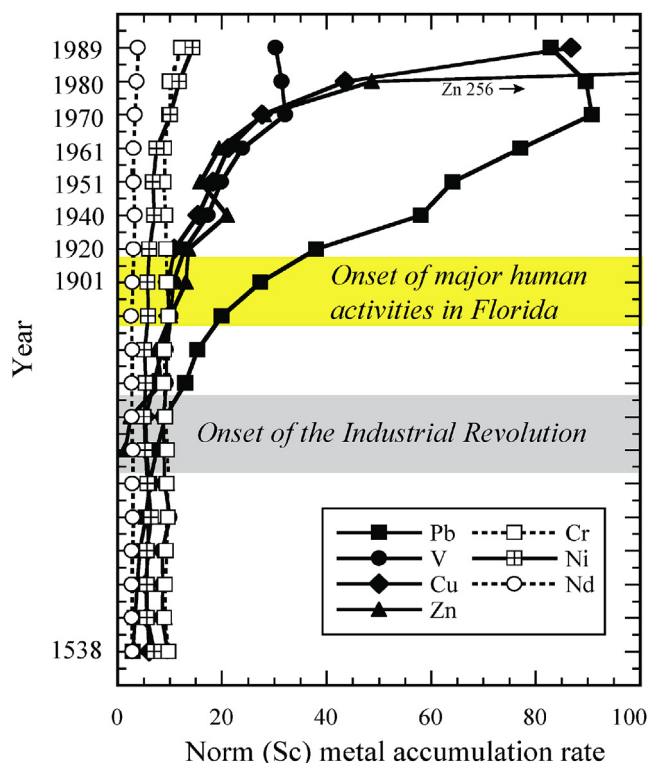


Fig. 2. Change in the atmospheric metal fluxes during the last 500 years in the Florida environment, data and figure from Kamenov et al. (2009). Metal fluxes are normalized to Sc in order to account for changes in the dust deposition rates, for more details see Kamenov et al. (2009). Note the drastic increase in the environmental fluxes of industrial metals such as Pb, V, Cu, and Zn during the 20th century due to industrial activities (see text for more details).

enhanced fluxes of industrial metals such as V, Cu, Zn, and Pb during the 20th century due to anthropogenic activities (e.g., Palanques et al., 2008; Kamenov et al., 2009). Furthermore, the peat record from Florida also shows elevated flux of Ni and Cr in modern times (Fig. 2). Therefore, the elevated concentrations of Ni, Cr, Cu, Zn, and Pb observed in the modern enamel, compared to the archaeological samples analyzed in this work (Table 1), must be a result of higher human exposure to these metals at present compared to pre-industrial times. Furthermore, regardless of localized Pb usage in the past (Montgomery et al., 2010; Price et al., 2017), the overall higher Pb content and specific Pb isotopic signatures in modern versus archaeological human skeletal material is widely recognized, attributable to enhanced modern human exposure to anthropogenic Pb (e.g., Patterson et al., 1991; Åberg et al., 1998; Budd et al., 2000; Kamenov and Gulson, 2014).

The enhanced metal exposure in modern times can be attributed to two major sources. One source is the overall environmental metal pollution caused by modern industrial activities (Nriagu and Pacyna, 1988; Rauch and Pacyna, 2009). Another source of trace metals to modern humans is water, food, supplements, and fortified food items with added minerals. For example, Cu, Fe, or Pb water supply pipes contribute to the modern human exposure to these metals (Clark, 2018). Furthermore, processed food items for children, including cereals and infant/toddler formulas, are often fortified with metals such as Ca, Mg, Fe, Mn, Cu, and Zn (e.g., <http://www.parentschoiceformula.com/infant-formula.aspx>). Similarly, conventional over-the-counter multivitamin supplements contain a number of essential and probably essential metals, including Ca, Mg, Cr, Cu, Zn, Fe, and Mn. Therefore, the observed overall higher Ni, Cr, Cu, Zn, and Pb concentrations in modern enamel can be attributed to elevated exposure of modern humans to these metals, from the overall enhanced environmental exposure due to industrial activities (Fig. 2) and/or from modern water,

food, or supplements.

4.2. Evidence for alteration of archaeological enamel based on V, Mn, Fe, La, Ce, Nd, Dy, Yb, Th, and U concentrations

4.2.1. V, Mn, and Fe

The same factors that control modern enamel chemistry (crystal structure, human physiology and nutrition, and metal exposure) will control the elemental composition of archaeological enamel. In addition, however, post-mortem diagenetic changes can also play a significant role in the observed present-day trace element composition of archaeological enamel. Trace element incorporation during bone diagenesis includes a complex range of processes, starting with very early adsorption mediated by organic (e.g., humic acids) and/or inorganic (e.g., Fe-Mn oxyhydroxides) agents, to advanced diagenesis with major chemical changes, to virtually complete “fossilization” associated with intense chemical and structural changes (e.g., Reynard and Balter, 2014).

The archaeological enamel data overall show lower concentrations of industrial metals such as Ni, Cr, Cu, Zn, and Pb (Table 1), in agreement with the above argument for overall lower metal exposure during pre-industrial times. As can be observed in Fig. 2, natural archives show much lower fluxes of the above metals before modern industrial times. However, regardless of the much lower V fluxes prior to the 20th century (Fig. 2), V is significantly higher in the archaeological enamel (Table 1, Figs. 1, 3 and 4). Crystal structure and human physiology is expected to be identical for modern and archaeological individuals, and also the overall environmental V exposure was much lower in the past (Fig. 2). Furthermore, modern diet is typically higher in V, even when compared to a diet consisting of seafood. For example, fish and shellfish V content range from 5 to 28 ng/g, which is similar or lower compared to common modern food items such as white bread (20 ng/g V), eggs (21 ng/g V), chicken (22 ng/g V), and cereal (93 ng/g V) (Myron et al., 1977). Cereal, in particular, may be consumed in greater quantities by modern children during the period of enamel formation.

As can be seen in Figs. 3 and 4, a number of archaeological samples show elevated C/MTC for V. Based on the higher V fluxes at present (Fig. 2) and overall dietary preferences, modern human enamel should show higher V concentrations when compared to archaeological non-altered enamel. The estimated MTC for V = 0.11 ppm (Table 1) is

below the maximum V = 0.3 ppm reported by Curzon et al. (1975) for deciduous teeth. However, the samples analyzed by Curzon et al. (1975) were all deciduous teeth, and were analyzed using older analytical instrumentation. As a result, these data are not directly comparable to the permanent tooth enamel data analyzed in this work. As observed in Figs. 1, 3 and 4, a number of archaeological samples show from several-fold higher to over a magnitude higher V, suggesting that post-mortem diagenetic changes, rather than in-vivo metal exposure, most likely caused the higher V concentrations. Recent work by Grimstead et al. (2018) identified elevated U and V concentrations in diagenetically altered areas of archaeological animal bone. Grimstead et al. (2018) suggested that V may substitute for Ca^{2+} as VO^{2+} or as VO_4^{3-} for PO_4^{3-} in the hydroxyapatite structure during diagenesis. Alternatively, it is possible that trace amounts of V are adsorbed at discrete sites in the enamel together with Fe-Mn oxyhydroxides, as discussed below.

Elevated Fe and Mn concentrations in archaeological bones are a result of precipitation of Fe^{3+} and Mn^{4+} oxyhydroxides during the post-mortem interval (Kohn et al., 1999; Lopez-Gonzales et al., 2006; Kuczmow et al., 2010; Rasmussen et al., 2013). Recently, Decrée et al. (2018) proposed that diagenetic Mn and/or Fe mineral phases, including todorokite (Mn-oxyhydroxide), are often present as discrete nano-scale coatings in fossil animal bones and can adsorb a number of trace metals, including V as VO^{2+} . Earlier, Kohn et al. (1999) identified sub-micron sized clays and Fe-Mn oxides in fossil animal dentine tubules. Furthermore, in situ transmission electron microscopy (TEM) showed presence of secondary manganite $\text{MnO}(\text{OH})$ and Fe-Mn oxyhydroxides in the fossil animal enamel (Kohn et al., 1999). Therefore, it is very likely that discrete, sub-micron Fe-Mn oxyhydroxides may become incorporated on the surface and/or along mineral grain boundaries, and/or in pores or cracks of the human enamel in particular burial environments.

It is expected that Mn and Fe will be higher in modern humans, given the wide availability of fortified foods particularly in Europe and the USA, and the overall much higher environmental exposure to industrial metals at present. Furthermore, Mn has been used as a gasoline additive in MMT (Methylcyclopentadienyl Manganese Tricarbonyl) since the 1970s (www.wikipedia.org). The archaeological samples analyzed in this work were carefully cleaned of any visible surficial alterations using a dental drill and were evaluated under a microscope

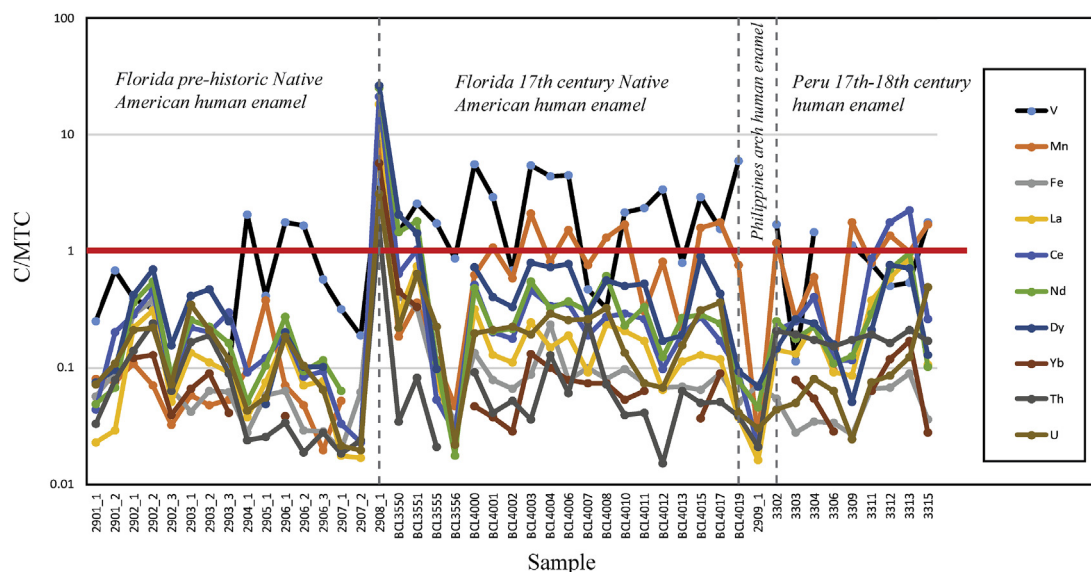


Fig. 3. Concentration (C) of V, Mn, Fe, La, Ce, Nd, Dy, Yb, Th, and U normalized to Maximum Threshold Concentration (MTC). The red line C/MTC = 1 represents the threshold value above which the archaeological enamel samples show diagenetic alteration. Note the highly enriched C/MTC signal in sample 2908_1 suggesting advanced diagenetic alteration, see the text for more details. Data from this work and Lofaro (2016). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

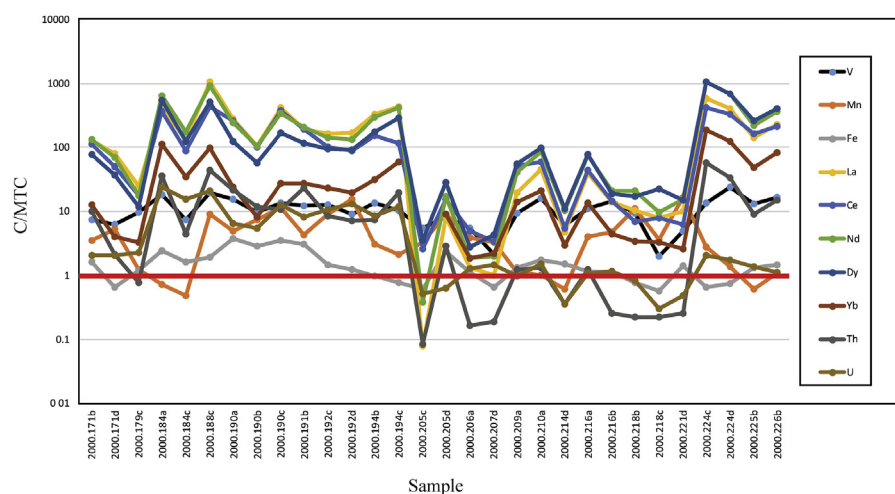


Fig. 4. Concentrations of V, Mn, Fe, La, Ce, Nd, Dy, Yb, Th, and U normalized to MTC (C/MTC) for a set of mid-Holocene archaeological enamel samples from Cis-Baikal region, Siberia (archaeological enamel data from Scharlotta and Weber, 2014). The red line C/MTC = 1 represents the threshold value above which the archaeological samples show diagenetic alteration. Note that most of the samples show highly enriched C/MTC indicating advanced diagenetic alteration (see text for more details). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

before dissolution. Therefore, the observed Mn and Fe above the MTC (Figs. 1, 3 and 4) in some of the archaeological samples suggest a chemical change due to diagenesis, likely a result of discrete Fe and/or Mn oxyhydroxides precipitation in micro-pores, tunnels, cracks, and/or along mineral grain boundaries. Overall, samples show more elevated Mn than Fe (Figs. 3 and 4). Separation of Mn from Fe in the burial environment can be a result of sequential precipitation during variable redox conditions and/or biomineralization of Mn oxides. Buried animal bones rapidly develop micron-sized tunneling features, a result of microbial attack in the post-mortem interval (Jans et al., 2004). Microbial and fungal soil communities readily oxidize Mn^{2+} to Mn^{4+} and so precipitate Mn-oxides and oxyhydroxides (Carmichael et al., 2013). Most samples showing elevated Mn also show elevated V, and the latter is also important for the development of bacteria, suggesting that biomineralization may play an important role in the early adsorption-dominated stages of the enamel diagenesis.

4.2.2. REE, U, and Th

REE and U are well-established indicators for diagenetic change in bones (e.g., Trueman et al., 2004; Reynard and Balter, 2014). REE in particular, increase in concentrations several-fold in a matter of months to years in animal bones during the postmortem interval (Trueman et al., 2004; Reynard and Balter, 2014). Furthermore, in contrast to traditional industrial metals, Nd (one of the REE) flux in the environment has been virtually constant for the last 500 years (Fig. 2). Given the overall constant REE flux in the environment, their non-essential role in human physiology, modern and archaeological enamel should show a similar range of “in-vivo” concentrations. Therefore, the observed elevated concentration of an REE such as Nd above the MTC in some archaeological samples (Figs. 1, 3 and 4) can also be used as a powerful indicator for post-mortem diagenetic changes.

On Figs. 1, 3 and 4 we use the MTC for La, Ce, Nd, Dy, and Yb, as the rest of the REE showed very low concentrations in modern enamel, either below or close to the detection limit (Table 1 and on-line supplement). Recently reported high-precision ID-TIMS data show Nd concentration range from 0.0001 ppm to 0.021 ppm in modern human tooth enamel (Plomp et al., 2017). The proposed MTC for Nd (0.058 ppm, Table 1) is close to three times higher than the maximum high-precision ID-TIMS (Nd = 0.021 ppm) data. Therefore, the Nd MTC should cover the expected range for “in-vivo” Nd concentrations in human enamel. As can be observed in Fig. 3, only several samples show REE above the MTC in the archaeological samples analyzed in this study. Similarly, U and Th are below the MTC in our archaeological samples with the exception of sample 2908_1, which shows all of the proposed metal indicators for alteration above the MTC. In addition, the much more recent Native American samples from Florida (BCL3550 to

BCL4019) show overall higher C/MTC for V and Mn compared to the prehistoric Floridian samples (2901_1 to 2907_2) (Fig. 3). Therefore, in just a few hundred years enamel can develop signs of diagenetic changes. This shows that enamel samples even from a single location or burial context can show a wide range of preservation, ranging from pristine to strongly altered during diagenesis. Therefore, even very recent archaeological enamel samples must be screened for post-mortem chemical changes.

Based on the observed C/MTC relationships we can estimate that some of the samples most likely still preserve the original “in-vivo” chemical signal (e.g., 2901_1, 2901_2, 2902_2, etc.) as they plot below C/MTC = 1 (Fig. 3). Other samples analyzed in this work (e.g., 2904_1, BCL4003, 3309) have apparently undergone early low-grade alteration, as they show MTC above 1 only for Mn and/or V (Fig. 3). This is most likely due to early adsorption of metals mediated by organic and/or inorganic (e.g., Fe-Mn oxyhydroxides) agents. The elemental and isotope information for such samples, in particular for low abundance elements such as Pb, may be compromised to some degree and data should be approached with caution, as will be discussed below. The most advanced alteration in our data set is observed in sample 2908_1 (Figs. 1 and 3). The elevated concentrations of REE in 2908_1 indicate that the alteration in this sample has progressed beyond the early surficial adsorption-dominated stage. It is well-known that REE, Sr, and Pb readily substitute for Ca in the crystal structure of apatite (e.g., Pan and Fleet, 2002; Reynard and Balter, 2014). Sample 2908_1 shows over an order of magnitude increase in La, Ce, Nd, Dy (Fig. 1) and Pb (Table 1, maximum Pb concentration in archaeological enamel analyzed in this work was measured in 2908_1), indicating that the diagenetic changes in this sample have advanced to cation substitution of REE, Pb, and other metals for Ca in the apatite structure. As a result, the elemental and isotope information for this sample will be compromised and will not represent the true “in-vivo” signal.

In Fig. 4 we apply the MTC normalization to evaluate published trace elements data for human enamel samples from the Baikal region, Siberia (Scharlotta and Weber, 2014) for signs of diagenetic alteration. Comparison of Figs. 1 and 4 shows that the enrichment in the Baikal enamel samples is similar or higher than the enrichment in sample 2908_1. As can be seen on Fig. 4, virtually all of the metals identified as indicators for alteration are above the MTC in the Baikal samples, with exception of several samples showing Th and U below the MTC. Overall, the Baikal data show much higher enrichment compared to the archaeological samples analyzed in this work (compare Figs. 3 and 4). The highly elevated REE in Baikal samples, in some cases up to 3 orders of magnitude (Fig. 4), show advanced diagenesis with apparent cation substitution of REE and other metals for Ca in virtually all of these enamel samples. The elemental and isotope information in such

samples will be compromised and will not be truly representative of the “in-vivo” enamel signal.

4.3. Mass-balance consideration for alteration effects on Pb and Sr isotope data

Although MTC can be used as a quick test to evaluate if enamel samples have undergone some degree of alteration, it is difficult to quantitatively estimate the diagenetic effects on the Pb and Sr isotopic compositions. One of the reasons is that Pb is highly elevated at present due to anthropogenic pollution. This results in MTC Pb = 63, which is much higher than the Pb concentration in the archaeological samples (Table 1). In addition, Sr shows relatively similar concentrations in modern and archaeological enamel samples (Table 1). Therefore, using only Pb and Sr concentration data in comparison with modern enamel will not provide a quantitative estimate for what proportion, or if any, of Pb or Sr were added to the archaeological enamel during the diagenesis.

As discussed above, the early low-grade diagenetic alteration is most likely dominated by discrete adsorption mediated by organic and/or inorganic (e.g., Fe-Mn oxyhydroxides) agents. Fe-Mn oxyhydroxides are well-known to concentrate a number of trace metals, including Pb and Sr, at tens to hundreds to thousands of ppm concentrations (e.g., Axelsson et al., 2002; Brown et al., 2018; Decr  e et al., 2018). Therefore, even a minor adsorption of such oxyhydroxides may have significant influence on the Pb-Sr isotope results. A simple mass-balance calculation, assuming metal addition during precipitation of discrete Fe-Mn oxyhydroxides in the enamel, can be used to estimate if Pb can be significantly affected by such low-degrees of alteration. Axelsson et al. (2002) reported up to 860 ppm Pb concentration in Fe-Mn oxyhydroxide nodule Nod-A1 (USGS standard). Using this concentration for the high-end estimate shows that addition of 1 µg oxyhydroxides with 860 ppm Pb will add 0.00086 µg Pb to the enamel. If we use the average 0.6 ppm Pb reported in Table 1 for archaeological enamel, then a typical 30 mg enamel sample for Sr-Pb isotopes will contain 0.018 µg Pb. In this particular example the addition of 0.00086 µg Pb will result in about 5% additional diagenetic Pb. This amount of Pb, depending on the isotopic differences between the two end-members, can modify, although to minor extent, the original “in-vivo” Pb isotope signal. Furthermore, if the oxyhydroxides contain less Pb, on the order of 10s of ppm, then this adsorption effect on the Pb isotopes will be virtually negligible. For example, if we use 86 ppm Pb (10x less than Nod-A1), which is in the range of freshwater Fe-Mn oxyhydroxides (e.g., Brown et al., 2018) then the diagenetically added Pb will be only 0.5%, assuming the same conditions as the above example. However, most pre-historical samples show Pb below 0.1 ppm (on-line supplement) and in such cases diagenetic Pb effect will be much more pronounced. For example, assuming sample with “in-vivo” Pb = 0.05 ppm, then the same simple mass-balance calculation of addition of 1 µg oxyhydroxides containing 860 ppm Pb will result in over 40% diagenetic Pb, or 4% assuming 86 ppm Pb. Therefore, enamel samples with low initial “in-vivo” Pb content will be much more susceptible to isotopic changes due to diagenetic Pb, unless both the “in-vivo” and diagenetic Pb have identical isotopic signature.

In contrast, the same simple mass-balance consideration shows that addition of small amounts of Fe-Mn oxyhydroxides will not have a significant effect on the Sr isotopic composition due to the overall much higher “in-vivo” Sr content in the enamel. Using the same assumptions as before, and the high Sr = 1630 ppm in Fe-Mn nodule Nod-A1 (Axelsson et al., 2002), then 1 µg oxyhydroxides will result in addition of 0.0016 µg Sr to the enamel. However, on average archaeological enamel contains 188 ppm Sr (Table 1) and this means 5.6 µg in-vivo Sr in a typical 30 mg sample. Therefore, the addition of 0.0016 µg Sr during the diagenetic processes will add only around 0.03% external Sr. Such a small amount should not have a significant effect on the original “in-vivo” Sr isotope composition. However, once the alteration goes

beyond the low-grade adsorption stage and proceeds to advanced diagenesis accompanied with cation substitution of Pb and Sr for Ca in the apatite structure (manifested by orders of magnitude increase in the REE), then both Pb and Sr isotope results may well be compromised and not representative of the “in-vivo” signal.

5. Conclusions

The chemical information recorded in human enamel can provide information about nutrition, metal exposure, and diagenetic alteration. Modern human enamel shows elevated concentrations of Ni, Cr, Cu, Zn, and Pb, a result of overall higher human exposure to industrial metals at present compared to pre-industrial times. This correlates with enhanced metal fluxes in the modern environment. However, regardless of the overall lower metal exposure during pre-industrial times, a number of archaeological samples show elevated V, Mn, Fe, REE, Th, and U concentrations. Apart from Fe, none of the above metals have been widely used by humans in pre-industrial times, but all are readily incorporated in apatite during diagenesis. Therefore, the observed elevated V, Mn, Fe, REE, Th, and U concentrations show that some of the archaeological enamel samples have undergone chemical changes during diagenesis. Based on the observed concentrations in modern enamel we propose a set of Maximum Threshold Concentration (MTC) values that can be used to evaluate the extent of diagenesis in archaeological enamel samples. Concentrations above the MTC for any or all of the following elements, V, Mn, Fe, REE, Th, and U, indicate that the archaeological human enamel has undergone chemical changes during the post-mortem interval.

The observed concentration variations indicate a wide range of preservation states in archaeological enamel samples even in burials with similar age and context. Some of the enamel samples do not show detectable trace element signature for diagenetic alteration and likely preserve the original, “in-vivo” compositions. Others show indications for early adsorption-driven changes mediated by organic and/or inorganic agents, evident in their elevated Mn and/or V concentrations, and some show evidence for highly advanced diagenesis manifested by orders of magnitude increase in the REE and/or U content. Simple mass-balance considerations show that although the early low degrees of alteration are unlikely to change the “in-vivo” Sr isotopic composition, the Pb isotopic composition can be affected to some degree. In contrast, the enamel samples showing advanced stage of alteration accompanied by cation substitution in the apatite structure, manifested by over a magnitude increase in the REE above the MTC, are chemically compromised. In such samples both Pb and Sr isotopes will be likely altered to different degrees depending on the initial “in-vivo” concentrations, but this will be evident isotopically only if there is enough isotopic contrast between the “in-vivo” and the burial environment isotopes. If there are no significant isotope differences then all samples will appear isotopically “local,” regardless of the alteration state. The results of this work show that, when possible, it is best to test all archaeological tooth enamel for diagenetic changes prior isotopic analyses, because a range of alteration stages may be observed even in samples from the same burial context.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jas.2018.09.002>.

References

- Åberg, G., Fosse, G., Stray, H., 1998. Man, nutrition and mobility: a comparison of teeth and bone from the medieval era and the present day from Pb and Sr isotopes. *Sci. Total Environ.* 224, 109–119.
- Austin, C., Smith, T.M., Bradman, A., Hinde, K., Joannes-Boyau, R., Bishop, D., Hare, D.J., Doble, P., Eskenazi, B., Arora, M., 2013. Barium distributions in teeth reveal early-life dietary transitions in primates. *Nature* 498, 216–219.
- Axelsson, M.D., Rodushkin, I., Ingri, J., Ohlander, B., 2002. Multielemental analysis of Mn-Fe nodules by ICP-MS: optimization of analytical method. *Analyst* (Cambridge, U.K.) 127, 76–82.
- Ayliffe, L.K., Chivas, A.R., Leakey, M.G., 1994. The retention of primary oxygen isotope compositions of fossil elephant skeletal phosphate. *Geochim. Cosmochim. Acta* 58, 5291–5298.
- Brown, A.L., Martin, J.B., Kamenov, G.D., Ezell, J.E., Sreaton, E.J., Gulley, J., Spellman, P., 2018. Trace metal cycling in karst aquifers subject to periodic river water intrusion. *Chem. Geol.* <https://doi.org/10.1016/j.chemgeo.2018.05.020>.
- Budd, P., Montgomery, J., Evans, J., Barreiro, B., 2000. Human tooth enamel as a record of the comparative lead exposure of prehistoric and modern people. *Sci. Total Environ.* 263, 1–10.
- Budd, P., Montgomery, J., Evans, J., Trickett, M., 2004. Human lead exposure in England from approximately 5500 BP to the 16th century AD. *Sci. Total Environ.* 318, 45–58.
- Carmichael, M.J., Carmichael, S.K., Santelli, C.M., Strom, A., Brauer, S.L., 2013. Mn(II)-ferromanganese deposits in caves of the upper Tennessee river basin. *Geomicrobiol. J.* 30, 779–800.
- Chiaradia, M., Galloway, A., Todt, W., 2003. Different contamination styles of prehistoric human teeth at a Swiss necropolis (Sion, Valais) inferred from lead and strontium isotopes. *Appl. Geochem.* 18, 353–370.
- Clark, A., 2018. *The Poisoned City: Flint's Water and the American Urban Tragedy*. Metropolitan Books, New York 320pp.
- Curzon, M.E., Losee, F.L., Macalister, A.D., 1975. Trace elements in the enamel of teeth from New Zealand and the USA. *New Zealand Dental Journal* 71, 80–83.
- Decr e, S., Herwartz, D., Mercadier, J., Ismael, M., de Buffr nil, V., Leduc, T., Lambert, O., 2018. The post-mortem history of a bone revealed by its trace element signature: the case of a fossil whale rostrum. *Chem. Geol.* 477, 137–150.
- Giovias, C.M., Kamenov, G.D., Fitzpatrick, S.M., Krigbaum, J., 2016. Sr and Pb isotopic investigation of mammal introductions: pre-columbian zoogeographic records from the lesser antilles, west Indies. *J. Archaeol. Sci.* 69, 39–43.
- Grimstead, D.N., Clark, A.E., Rezac, A., 2018. Uranium and vanadium concentrations as a trace element method for identifying diagenetically altered bone in the inorganic phase. *J. Archaeol. Meth. Theor.* 25, 689–704.
- Gulson, B.L., Gillings, B.R., 1997. Lead exchange in teeth and bone—a pilot study using stable lead isotopes. *Environ. Health Perspect.* 105, 820–824.
- Hinz, E.A., Kohn, M.J., 2010. The effect of tissue structure and soil chemistry on trace element uptake in fossils. *Geochim. Cosmochim. Acta* 74, 3213–3231.
- Jans, M.M.E., Nielsen-Marsh, C.M., Smith, C.I., Collins, M.J., Kars, H., 2004. Characterization of microbial attack on archaeological bone. *J. Archaeol. Sci.* 31, 87–95.
- Kamenov, G.D., Gulson, B.L., 2014. The Pb isotopic record of historical to modern human lead exposure. *Sci. Total Environ.* 490, 861–870.
- Kamenov, G.D., Brenner, M., Tucker, J.L., 2009. Anthropogenic vs natural control on trace element and Sr-Nd-Pb isotope record in peat sediments of southeast Florida, USA. *Geochim. Cosmochim. Acta* 73, 3549–3567.
- Keenan, S.W., Engel, A.S., 2017. Early diagenesis and recrystallization of bone. *Geochim. Cosmochim. Acta* 196, 209–223.
- Kohn, M.J., Moses, R.J., 2013. Trace element diffusivities in bone rule out simple diffusive uptake during fossilization but explain in vivo uptake and release. *Proc. Natl. Acad. Sci. Unit. States Am.* 110, 419–424.
- Kohn, M.J., Schoeninger, M.J., Barker, W.W., 1999. Altered states: effects of diagenesis on fossil tooth chemistry. *Geochim. Cosmochim. Acta* 63, 2737–2747.
- Kuczmow, A., Cukrowska, E., Stachniuk, A., Gaweda, R., Mroczka, R., Paszkowicz, W., Skrzypiec, K., Falkenberg, R., Backwell, L., 2010. Investigation of chemical changes in bone material from South Africa fossil hominid deposits. *J. Archaeol. Sci.* 37, 107–114.
- Lee-Thorp, J.A., van der Merwe, N.J., 1991. Aspects of the chemistry of modern and fossil biological apatites. *J. Archaeol. Sci.* 18, 343–354.
- Lofaro, E.M., 2016. *Ad Majorem Dei Gloriam: an Isotopic Investigation of Indigenous Lifeways in a Jesuit Church from Early Colonial Huamanga (Ayacucho), Peru*. PhD Dissertation. Department of Anthropology, University of Florida.
- Lopez-Gonzalez, F., Grandal-d'Anglade, A., Vidal-Romani, J.R., 2006. Deciphering bone depositional sequences in caves through the study of manganese coatings. *J. Archaeol. Sci.* 33, 707–717.
- MacFadden, B.J., DeSantis, L.R., Labs-Hochstein, J., Kamenov, G.D., 2010. Physical properties, geochemistry, and diagenesis of xenarthran teeth: prospects for interpreting the paleoecology of extinct species. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 291, 180–189.
- Montgomery, J., Evans, J., Chenery, S., Pashley, V., Killgrove, K., 2010. “Gleaming, white and deadly”: using lead to track human exposure and geographic origins in the Roman period in Britain. In: Eckardt, H. (Ed.), *Roman Diasporas: Archaeological Approaches to Mobility and Diversity in the Roman Empire*, *Journal of Roman Archaeology, Supplement*. vol. 78. pp. 199–226 11.
- Myron, D.R., Givand, S.H., Nielsen, F.H., 1977. Vanadium content of selected foods as determined by flameless atomic absorption spectroscopy. *J. Agric. Food Chem.* 25, 297–300.
- Nelson, B.J., Deniro, M.J., Schoeninger, M.J., De Paolo, D.J., Hare, P.E., 1986. Effects of diagenesis on strontium, carbon, nitrogen and oxygen concentration and isotopic composition of bone. *Geochim. Cosmochim. Acta* 50, 1941–1949.
- Nriagu, J.O., Pacyna, J.M., 1988. Quantitative assessment of worldwide contamination of air, water and soils by trace elements. *Nature* 333, 134–139.
- Palanques, A., Masque, P., Puig, P., Sanchez-Cabeza, J.A., Frignani, M., Alvisi, F., 2008. Anthropogenic trace metals in the sedimentary record of the Llobregat continental shelf and adjacent Foix Submarine Canyon (northwestern Mediterranean). *Mar. Geol.* 248, 213–227.
- Pan, Y., Fleet, M.E., 2002. Composition of the apatite-group minerals: substitution mechanisms and controlling factors. In: Kohn, M.J., Rakovan, J., Hughes, J.M. (Eds.), *Phosphates: Geochemical, Geobiological and Materials Importance. Reviews in Mineralogy & Geochemistry*. vol. 48. pp. 13–49.
- Patterson, C., Ericson, J., Manea-Krichten, M., Shirahata, H., 1991. Natural skeletal levels of lead in *Homo sapiens sapiens* uncontaminated by technological lead. *Sci. Total Environ.* 107, 205–236.
- Plomp, E., von Holstein, C.C., Koornneef, J.M., Smets, R.J., Font, L., Baart, J.A., Forouzanfar, T., Davies, G.R., 2017. TIMS analysis of neodymium isotopes in human tooth enamel using $10^{13} \Omega$ amplifiers. *J. Anal. At. Spectrom.* 32, 2391–2400. <https://doi.org/10.1039/c7ja00312a>.
- Price, T.D., Frei, R., Backstrom, Y., Frei, K.M., Ingvarsson-Sundstrom, A., 2017. Origins of inhabitants from the 16th century Sala (Sweden) silver mine cemetery – a lead isotope perspective. *J. Archaeol. Sci.* 80, 1–13.
- Rasmussen, K.L., Skytte, L., Pilek r, C., Lauritsen, A., Boldsen, J.L., Leth, P.M., Thomsen, P.O., 2013. The distribution of mercury and other trace elements in the bones of two human individuals from medieval Denmark – the chemical life history hypothesis. *Heritage Science* 23, 1–10. <https://doi.org/10.1186/2050-7445-1-10>.
- Rauch, J.N., Pacyna, J.M., 2009. Earth's global Ag, Al, Cr, Cu, Fe, Ni, Pb, and Zn cycles. *Global Biogeochemical Cycles* 23, GB2001. <https://doi.org/10.1029/2008GB003376>.
- Reid, D.J., Dean, M.C., 2006. Variations in modern human enamel formation times. *J. Hum. Evol.* 50, 329–346.
- Reynard, B., Balter, V., 2014. Trace elements and their isotopes in bone and teeth: diet, environment, diagenesis, and dating of archaeological and paleontological samples. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* 416, 4–16.
- Scharlotta, I., Weber, A., 2014. Mobility of middle Holocene foragers in the Cis-Baikal region, Siberia: individual life history approach, strontium ratios, rare earth and trace elements. *Quat. Int.* 348, 37–65.
- Trueman, C.N., Tuross, N., 2002. Trace elements in recent and fossil bone. In: Kohn, M.J., Rakovan, J., Hughes, J.M. (Eds.), *Phosphates: Geochemical, Geobiological and Materials Importance. Reviews in Mineralogy & Geochemistry*. vol. 48. pp. 489–521.
- Trueman, C.N.G., Behrensmeier, A.K., Tuross, N., Weiner, S., 2004. Mineralogical and compositional changes in bones exposed on soil surfaces in Amboseli National Park, Kenya: diagenetic mechanisms and the role of sediment pore fluids. *J. Archaeol. Sci.* 31, 721–739.
- Turner, B.L., Kamenov, G.D., Kingston, J.D., Armelagos, G.J., 2009. Insights into immigration and social class at Machu Picchu, Peru, based on oxygen, strontium, and lead isotopic analyses. *J. Archaeol. Sci.* 36, 317–332.
- Valentine, B., Kamenov, G.D., Kenoyer, J.M., Shinde, V., Mushrif-Tripathy, V., Otarola-Castillo, E., Krigbaum, J., 2015. Evidence for patterns of selective urban migration in the Greater Indus Valley (2600–1900 BC): a lead and strontium isotope mortuary analysis. *PLoS One* 10, e0123103. <https://doi.org/10.1371/journal.pone.0123103>.